

Mark Busenitz

Software Engineer

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Cross-platform mobile and game developer with over 15 years of experience in building SDKs, games and tools. Demonstrated ability to design robust APIs, debug low-level native implementations, and deliver high-quality software across multiple platforms. Individually researched, implemented, documented and led integration of features from ideas to production.

Work Experience

Senior Software Engineer | Mobile SDK Development

Glu mobile (acquired by EA) | June 2017 - September 2023

- Lead developer of centralized Advertising SDK supporting rewarded video, native ad, offerwall and interstitial ad formats. Out-of-the-box features including ad suppression rules, rule-based-frequency capping and personalized rewards.
- Core contributor of a suite of SDKs included in EA Mobile portfolio games such as FC Mobile, SimCity BuildIt, Design Home, Golf Clash, etc.
 - Developed SDKs for Unity, Unreal and custom engines on iOS and android.
 - Worked closely with QA to ensure testability of new features.
 - Assisted game teams with SDK integration and verification.
 - Reported and verified bugs with external teams.
 - Maintained and documented 3rd party dependencies, including network endpoints used.
 - Built automation tools for SDK testing.
 - Improved firebase crash reporting by including SDK and game logs.
- Lead developer of growth initiatives with EA Mobile.
 - Ideated and implemented an in-house playable ads platform for user acquisition, providing cost savings for the game teams.
 - Worked with the product team to develop cross-promotion functionality with attribution tracking.

Software Engineer | Desktop app development

Blurb | December 2013 - May 2017

- Developed and maintained the flagship product using the Qt framework.
- Assisted in urgent customer issues at moment's notice.
- Helped reduce the crash rate from 8% to 1.5% by consistently fixing top offenders as tracked by a crash database.
- Sped up the teams development by introducing assertions.
- Reduced upload failures by implementing a chunked uploader.
- Included a static analyzer check to the build chain.
- Reduced build time by 20% by eliminating building of 3rd party tests.
- Wrote script to automate patching and building of the Qt binaries.

Software Engineer | Desktop applications

OPSWAT | May 2013 - August - 2013

- Wrote test automation using Perl.
- Verified release candidates using virtual machines.
- Documented manual test cases.
- Fixed reported bugs.

Software Engineer | Game developer

Tricky Software | September 2007 - March 2013

- Developed high-profile iOS titles such as Spore Origins and Boggle.
- Wrote game prototypes for clients including EA, Disney and Mattel.
- Optimized in-house game engine by eliminating redundant calculations and researched possibilities of new iOS

features such as dual monitor games.

- Helped the company grow from 1 to 10 employees by leading new engineers.
- Convinced stakeholders to adopt modern C++ instead of C with classes.
- Learned how to scale a shared code base targeting multiple platforms.
- Wrote Photoshop exporter plugin using JavaScript.

Skills

Languages

Proficient: C++, Objective-C, Java, WGSL

Familiar: C#, Kotlin, Python, Swift, JavaScript, HTML, CSS

Development Tools

- Graphics: WebGPU
- Web: NodeJS, Deno, Emscripten
- Build Systems: cmake, gradle, xcodebuild
- Network Inspection: Charles Proxy, mitmproxy

Personal Projects

mucbuc.com

Procedural animation on website <https://mucbuc.com>:

- Use C++ API for WebGPU
- Generate signed distance field for each glyph on GPU
- Automatically adjust contours to frame specified DOM elements
- Dynamic reflection of light source controlled by mouse

This demonstrates my experience with C++, emscripten, WebGPU and WGSL compute and fragment shaders. Best viewed in Chrome for full WebGPU animation.

MacOS apps

MacOS applications published on the app store using following tech:

- google dawn WebGPU for graphics
- HTML/CSS/JS inside of webview for UI
- Firebase Crashlytics
- C++ and Objective C

These projects demonstrate my skill level at general app and shader development. Due to Apple's developer fees they're not available on the App Store anymore. Download links below (requires Control-click to open as they're not App Store downloads).

FactorPuzzle

Simple puzzle game in which player needs to determine a number displayed as prime factors:

<https://github.com/mucbuc/website/raw/refs/heads/main/resources/FactorPuzzle.zip>

Morph

Desktop clock app with animating text: <https://github.com/mucbuc/website/raw/refs/heads/main/resources/morph.zip>

AutoStrings

Toy instrument with visual effects for played strings:

<https://github.com/mucbuc/website/raw/refs/heads/main/resources/AutoStrings.zip>